



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2015

**HIGHER SCHOOL CERTIFICATE
TRIAL PAPER**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Leave your answers in the simplest exact form, unless otherwise stated.
- Start each **NEW** question in a separate answer booklet.

Total Marks – 70

Section I

Pages 2–4

10 Marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section.

Section II

Pages 6–11

60 marks

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Examiners: *R. Elliot & J. Chen*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Section I

10 marks

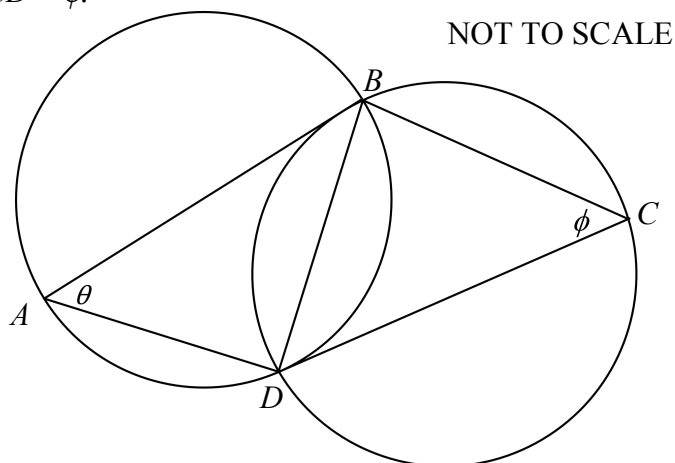
Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 The roots of $3x^3 - 2x^2 + x - 1 = 0$ are α , β and γ .
What is the value of $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$?
- (A) $-\frac{1}{9}$
(B) $-\frac{2}{9}$
(C) 1
(D) $\frac{2}{9}$
- 2 What is the minimum value of $\sqrt{7}\sin x - 3\cos x$?
- (A) -2
(B) -4
(B) -16
(D) $\sqrt{7} - 3$
- 3 What is the domain and range of $y = \sin^{-1}\left(\frac{2x}{5}\right)$?
- (A) Domain: $-1 \leq x \leq 1$; Range: $-\pi \leq y \leq \pi$
(B) Domain: $-1 \leq x \leq 1$; Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
(C) Domain: $-\frac{5}{2} \leq x \leq \frac{5}{2}$; Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
(D) Domain: $-\frac{5}{2} \leq x \leq \frac{5}{2}$; Range: $-\pi \leq y \leq \pi$
- 4 Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$.
- (A) 0
(B) $\frac{2}{3}$
(C) 1
(D) $\frac{3}{2}$

- 5 In the diagram below, AB is a tangent to the circle BCD .
Also, CD is a tangent to the circle ABD .
 $\angle BAD = \theta$ and $\angle BCD = \phi$.



Which of the following is a true statement?

- (A) $\triangle ABD \equiv \triangle BDC$
- (B) $ABCD$ is a cyclic quadrilateral
- (C) $\triangle ABD \parallel \triangle BDC$
- (D) $AB \parallel CD$
- 6 A particle moves in simple harmonic motion so that its velocity, v , is given by

$$v^2 = 6 - x - x^2.$$

Between which two points does it oscillate?

- (A) $x = 6$ and $x = 3$
- (B) $x = -2$ and $x = 3$
- (C) $x = 1$ and $x = 2$
- (D) $x = 2$ and $x = -3$
- 7 Which of the following is an expression for $\int \cos^3 x \sin x \, dx$?

- (A) $-\cos^4 x + c$
- (B) $-\frac{1}{4}\cos^4 x + c$
- (C) $\cos^4 x + c$
- (D) $\frac{1}{4}\cos^4 x + c$

- 8 Which of the following is the correct expression for the inverse of $f(x) = e^{1-2x}$?
- (A) $f^{-1}(x) = -2e^{1-2x}$
- (B) $f^{-1}(x) = -\frac{1}{2}e^{1-2x}$
- (C) $f^{-1}(x) = -\frac{1}{2}\log_e(1-2x)$
- (D) $f^{-1}(x) = \frac{1}{2}(1 - \log_e x)$
- 9 Three Mathematics study guides, four Mathematics textbooks and five exercise books are randomly placed along a bookshelf. What is the probability that the Mathematics textbooks are all next to each other?
- (A) $\frac{4!}{12!}$
- (B) $\frac{9!}{12!}$
- (C) $\frac{4!3!5!}{12!}$
- (D) $\frac{4!9!}{12!}$
- 10 A particle moves on the x -axis with velocity v m/s, such that $v^2 = 16x - x^2$. Which of the following is the particle's maximum speed and the position of where this maximum speed occurs?
- (A) Maximum speed = 16 m/s at $x = 0$
- (B) Maximum speed = 8 m/s at $x = -8$
- (C) Maximum speed = -8 m/s at $x = 8$
- (D) Maximum speed = 8 m/s at $x = 8$

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Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a NEW writing booklet. Extra pages are available

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks) Start a NEW Writing Booklet

- (a) Differentiate $\sin^{-1}(\log_e x)$. **1**
- (b) Find $\int \frac{1}{\sqrt{4-9x^2}} dx$. **1**
- (c) (i) Simplify $\sin(A+B) + \sin(A-B)$. **1**
- (ii) Hence, evaluate $\int_0^{\frac{\pi}{6}} \sin 3x \cos x \, dx$. **2**
- (d) The point $P(6p, 3p^2)$ is a point on the parabola $x^2 = 12y$.
- (i) Find the equation of the tangent at P . **2**
- (ii) The tangent at P cuts the y -axis at B . **3**
The point A divides PB internally in the ratio 1 : 2.
Find the locus of the point A as P varies.
- (e) A piece of meat at temperature $T^\circ \text{C}$ is placed in an oven, which has a constant temperature of $H^\circ \text{C}$.
The rate at which the temperature of the meat warms is given by
- $$\frac{dT}{dt} = -K(T - H),$$
- where t is in minutes and for some positive constant K .
- (i) Show that $T = H + Be^{-Kt}$, where B is a constant, is a solution of the differential equation above. **1**
- (ii) If the meat warms from 10°C to 50°C in the oven, which has a constant temperature of 180°C , in 30 minutes, find the value of K . **2**
- (iii) How long will it take the meat to get to a temperature of 150°C ? **2**
Express your answer correct to the nearest minute.

Question 12 (15 Marks) Start a NEW Writing Booklet

(a) (i) Solve $\cos x - \sqrt{3} \sin x = 1$ for $0 \leq x \leq 2\pi$. **2**

(ii) Hence, or otherwise, find a general solution to $\cos x - \sqrt{3} \sin x = 1$. **1**

(b) (i) On the same set of axes sketch the graphs of $y = \cos 2x$ and $y = \frac{x+1}{3}$ **2**

(ii) Use the graph to determine the number of solutions to the equation **1**

$$3\cos 2x = x + 1$$

(iii) One solution of the equation $3\cos 2x = x + 1$ is close to 0.5. **3**
Use one application of Newton's Method to find another approximation, correct to 3 decimal places.

(c) Evaluate $\int_0^{\frac{\pi}{4}} \sin^2 2x \, dx$ **3**

(d) When x cm from the origin, the acceleration of a particle moving in a straight line is given by: **3**

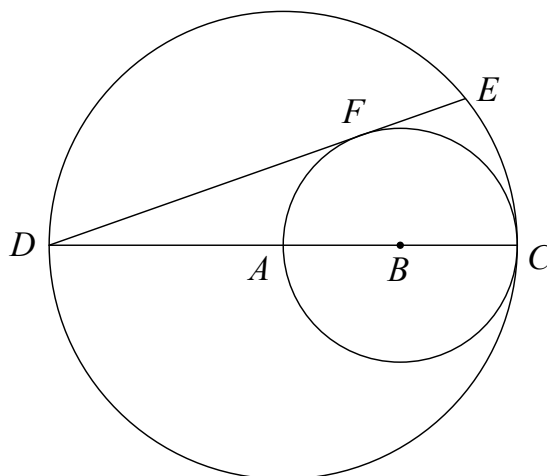
$$\frac{d^2x}{dt^2} = -\frac{5}{(x+2)^3}$$

It has an initial velocity of 2 cm/s at $x = 0$. If the velocity is V cm/s, find V in terms of x .

Question 13 (15 Marks) Start a NEW Writing Booklet

- (a) In the diagram below, DC is a diameter of the larger circle centred at A .
 AC is a diameter of the smaller circle centred at B .
 DE is tangent to the smaller circle at F and $DC = 12$.

Copy the diagram to your answer booklet.
Determine the length of DE .



- (b) (i) Simplify $k! + k \times k!$ **1**
- (ii) Prove, by mathematical induction, that **3**

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + n \times n! = (n+1)! - 1$$

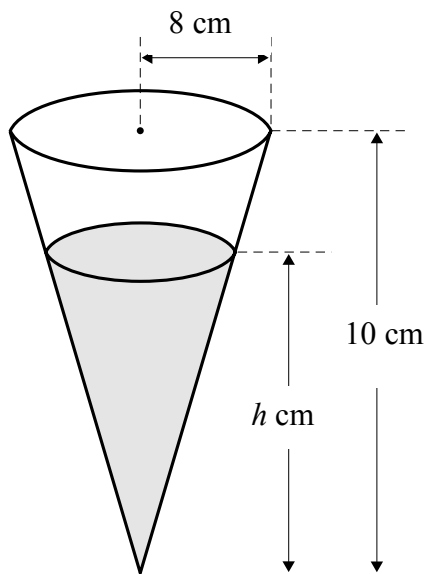
for all positive integers n .

- (c) (i) Using the substitution $x = 3 + 3\sin\theta$ find $\int \sqrt{x(6-x)} \, dx$ 4
- (ii) Let R be the region bounded by the curve $y = \sqrt[4]{x(6-x)}$ and the x -axis. 3
Find the volume of the solid of revolution generated by revolving R about the x -axis.

End of Question 13

Question 14 (15 Marks) Start a NEW Writing Booklet

(a)



The figure above shows an inverted conical cup with base radius 8 cm and height 10 cm.

Some water is poured into the cup at a constant rate of $\frac{2\pi}{5} \text{ cm}^3$ per minute.

Let the depth of the water be h cm at time t minutes.

Find the rate of change in the area of the water surface when $h = 4$

3

- (b) A particle is projected horizontally at 30 ms^{-1} from the top of a 100 m high wall. Assume that acceleration due to gravity is 10 ms^{-2} and that there is no air resistance.

The flight path of the particle is given by:

$$x = 30t, y = 100 - 5t^2 \text{ (Do NOT prove this)}$$

where t is the time in seconds after take-off.

- (i) Find the time taken for the particle to reach the ground.

1

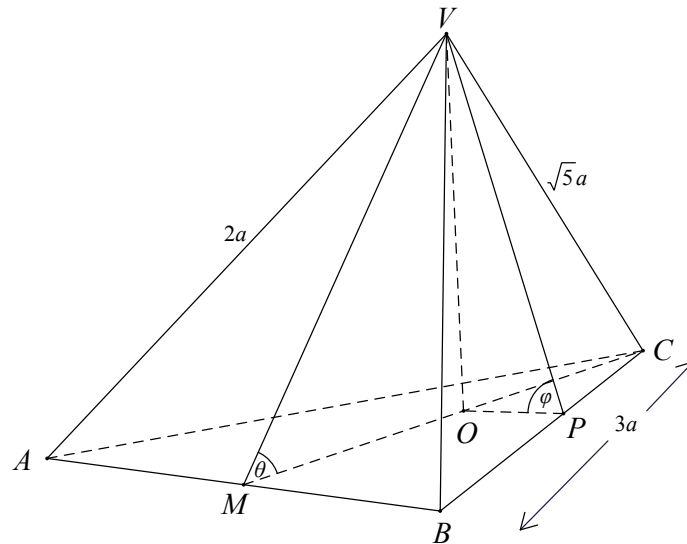
- (ii) Find the angle and speed at which the particle strikes the ground.

2

Question 14 continues on page 11

Question 14 (continued)

- (c) The diagram below shows a tetrahedron such that $VA = VB = AB = 2a$,
 $CA = CB = 3a$ and $VC = \sqrt{5}a$.
 O is the foot of the perpendicular from V to the base ABC .
 M is the midpoint of AB .
 P is a point on BC such that $BP = ra$ where $0 \leq r \leq 3$.
 $\angle VMC = \theta$ and $\angle VPO = \varphi$.



- (i) By considering $\triangle VMC$, show that $\cos \theta = \frac{\sqrt{6}}{4}$. 3
- (ii) Hence find the exact value of VO . 1
- (iii) Show that $VP^2 = \frac{1}{3}(3r^2 - 8r + 12)a^2$ 2
- (iv) Hence show that $\sin \varphi = \sqrt{\frac{45}{8(3r^2 - 8r + 12)}}$ 1
- (v) Hence, or otherwise, find the maximum value of φ as r varies. 2

End of paper



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Mathematics Extension 1

Sample Solutions

Question	Teacher
Q11	RB
Q12	BK
Q13	BD
Q14	PB

MC Answers

Q1	D
Q2	B
Q3	C
Q4	D
Q5	C
Q6	D
Q7	B
Q8	D
Q9	D
Q10	D

Section I 10 marks

- 1 The roots of $3x^3 - 2x^2 + x - 1 = 0$ are α , β and γ .
What is the value of $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$?

- (A) $-\frac{1}{9}$
(B) $-\frac{2}{9}$
(C) 1
(D) $\frac{2}{9}$

ANSWER: D

$$3x^3 - 2x^2 + x - 1 = 0$$

$$\alpha\beta\gamma = -\frac{d}{a} \qquad \alpha + \beta + \gamma = -\frac{b}{a}$$

$$= \frac{1}{3} \qquad \qquad \qquad = \frac{2}{3}$$

$$\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2 = \alpha\beta\gamma(\alpha + \beta + \gamma)$$

$$= \frac{1}{3} \times \frac{2}{3}$$

$$= \frac{2}{9}$$

- 2 What is the minimum value of $\sqrt{7}\sin x - 3\cos x$?

- (A) -2
(B) -4
(B) -16
(D) $\sqrt{7} - 3$

ANSWER: B

$$\sqrt{7}\sin x - 3\cos x$$

$$r = \sqrt{(\sqrt{7})^2 + 3^2}$$

$$= \sqrt{7+9}$$

$$= 4$$

$$\text{Let } \sqrt{7}\sin x - 3\cos x = r\sin(\theta - \alpha)$$

$$= 4\sin(\theta - \alpha)$$

No matter what the value of α

$$-1 \leq \sin(x - \alpha) \leq 1$$

$$-4 \leq 4\sin(x - \alpha) \leq 4$$

Therefore the minimum value is $x = -4$.

- 3 What is the domain and range of $y = \sin^{-1}\left(\frac{2x}{5}\right)$?
- (A) Domain: $-1 \leq x \leq 1$; Range: $-\pi \leq y \leq \pi$
- (B) Domain: $-1 \leq x \leq 1$; Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
- (C) Domain: $-\frac{5}{2} \leq x \leq \frac{5}{2}$; Range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
- (D) Domain: $-\frac{5}{2} \leq x \leq \frac{5}{2}$; Range: $-\pi \leq y \leq \pi$

ANSWER: C

$$y = \sin^{-1}\left(\frac{2x}{5}\right) \Rightarrow \sin y = \left(\frac{2x}{5}\right)$$

$$\text{Domain: } -1 \leq \sin y \leq 1$$

$$-1 \leq \frac{2x}{5} \leq 1$$

$$-\frac{5}{2} \leq x \leq \frac{5}{2}$$

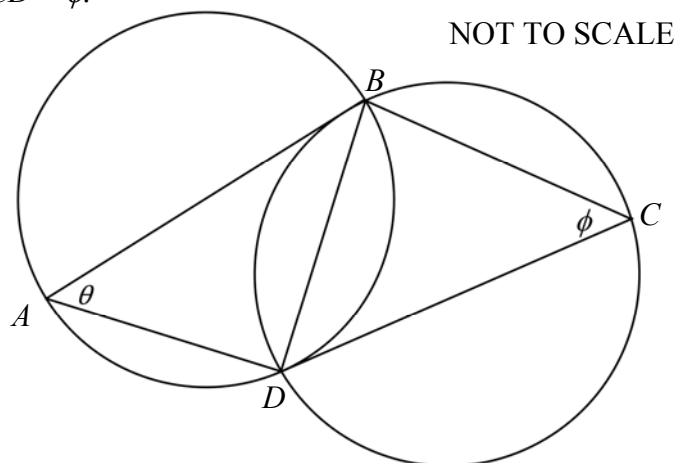
$$\text{Range: } -\frac{\pi}{2} \leq y \leq \frac{\pi}{2} \text{ as this is the range of } y = \sin^{-1} x$$

- 4 Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$.
- (A) 0
- (B) $\frac{2}{3}$
- (C) 1
- (D) $\frac{3}{2}$

ANSWER: D

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin 3x}{2x} &= \frac{3}{2} \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \times \frac{2}{3} \\ &= \frac{3}{2} \lim_{x \rightarrow 0} \frac{2 \sin 3x}{2 \times 3x} \\ &= \frac{3}{2} \times 1 \\ &= \frac{3}{2} \end{aligned}$$

- 5 In the diagram below, AB is a tangent to the circle BCD .
Also, CD is a tangent to the circle ABD .
 $\angle BAD = \theta$ and $\angle BCD = \phi$.



Which of the following is a true statement?

- (A) $\triangle ABD \equiv \triangle BDC$
(B) $ABCD$ is a cyclic quadrilateral
(C) $\triangle ABD \parallel \triangle BDC$
(D) $AB \parallel CD$

ANSWER: C

In $\triangle ABD$ and $\triangle DCB$:

$\angle DCB = \angle DBA$ (angle in the alternate segment)

ie $\angle DCB = \angle ABD$

$\angle BAD = \angle BDC$ (angle in the alternate segment)

ie $\angle BAD = \angle CDB$

BD is but not respective to angles.

Therefore $\triangle ABD \not\equiv \triangle DCB$

Hence, triangles are equiangular they are similar and $\triangle ABD \parallel \triangle DCB$

- 6 A particle moves in simple harmonic motion so that its velocity, v , is given by

$$v^2 = 6 - x - x^2.$$

Between which two points does it oscillate?

- (A) $x = 6$ and $x = 3$
(B) $x = -2$ and $x = 3$
(C) $x = 1$ and $x = 2$
(D) $x = 2$ and $x = -3$

ANSWER: D

$$v^2 = 6 - x - x^2$$

For the particle to reach its oscillation points $v = 0$.

$$v^2 = 6 - x - x^2$$

$$0 = 6 - x - x^2$$

$$0 = (3 + x)(2 - x)$$

$$\therefore x = -3 \text{ and } 2$$

7 Which of the following is an expression for $\int \cos^3 x \sin x \, dx$?

(A) $-\cos^4 x + c$

(B) $-\frac{1}{4}\cos^4 x + c$

(C) $\cos^4 x + c$

(D) $\frac{1}{4}\cos^4 x + c$

ANSWER: B

$\int \cos^3 x \sin x \, dx$, testing solutions:

$$\frac{d}{dx}(\cos^4 x) = 4\cos^3 x \times -\sin x$$

$$\frac{d}{dx} -\frac{1}{4}(\cos^4 x) = \cos^3 x \sin x$$

$$-\frac{1}{4}\cos^4 x = \int \cos^3 x \sin x \, dx$$

8 Which of the following is the correct expression for the inverse of $f(x) = e^{1-2x}$?

(A) $f^{-1}(x) = -2e^{1-2x}$

(B) $f^{-1}(x) = -\frac{1}{2}e^{1-2x}$

(C) $f^{-1}(x) = -\frac{1}{2}\log_e(1-2x)$

(D) $f^{-1}(x) = \frac{1}{2}(1 - \log_e x)$

ANSWER: D

Let $y = e^{1-2x}$

$$y = e^{1-2x}$$

$$\ln y = 1 - 2x$$

$$2x = 1 - \ln y$$

$$x = \frac{1}{2}(1 - \ln y)$$

$$\therefore f^{-1}(x) = \frac{1}{2}(1 - \ln y)$$

- 9 Three Mathematics study guides, four Mathematics textbooks and five exercise books are randomly placed along a bookshelf. What is the probability that the Mathematics textbooks are all next to each other?

(A) $\frac{4!}{12!}$

(B) $\frac{9!}{12!}$

(C) $\frac{4!3!5!}{12!}$

(D) $\frac{4!9!}{12!}$

ANSWER: D

Since there are 9 elements counting the textbooks as 1 element, hence these can be arranged in $9!$ ways. Also the textbooks can be arranged in $4!$ ways.

As there are 12 separate elements, the divisor for population can be counted in $12!$ ways.

Therefore, the probability is $\frac{9!4!}{12!}$

- 10 A particle moves on the x -axis with velocity v m/s, such that $v^2 = 16x - x^2$. Which of the following is the particle's maximum speed and the position of where this maximum speed occurs?

(A) Maximum speed = 16 m/s at $x = 0$

(B) Maximum speed = 8 m/s at $x = -8$

(C) Maximum speed = -8 m/s at $x = 8$

(D) Maximum speed = 8 m/s at $x = 8$

ANSWER: D

$$v^2 = 16x - x^2$$

$$\frac{1}{2}v^2 = 8x - \frac{x^2}{2}$$

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = 8 - \frac{2x}{2}$$

$$= 8 - x$$

$$\ddot{x} = 8 - x$$

When $\ddot{x} = 0$ the speed is the greatest so,

$$\ddot{x} = 8 - x$$

$$0 = 8 - x$$

$$x = 8$$

At $x = 8$,

$$v^2 = 16x - x^2$$

$$= 16(8) - (8)^2$$

$$= 64$$

$$v = \pm 8$$

As v , velocity can take positive and negative values, but the speed can only be positive, the maximum speed is 8 m/s.

(a) let $y = \sin^{-1}(\ln x)$

$$y' = \frac{1}{\sqrt{1-(\ln x)^2}} \times \frac{1}{x}$$

$$= \frac{1}{x\sqrt{1-(\ln x)^2}}$$

generally well answered but some students forgot the $\frac{1}{x}$.

① Others thought $(\ln x)^2 = 2\ln x$ or $\ln x^2$ No!

15

(b) $\int \frac{1 dx}{\sqrt{4-9x^2}} = \frac{1}{3} \sin^{-1}\left(\frac{3x}{2}\right) + C$ (1)

generally well done but some students left off the $\frac{1}{3}$.

(c) (i) $\sin(A+B) + \sin(A-B)$
 $\sin A \cos B + \cos A \sin B + \sin A \cos B - \cos A \sin B$

$= 2\sin A \cos B$ (1) this part, very well answered by 95% of students.

(ii) $\int_0^{\frac{\pi}{6}} \sin 3x \cos x dx$

$= \frac{1}{2} \int_0^{\frac{\pi}{6}} 2 \sin 3x \cos x dx$

$= \frac{1}{2} \int_0^{\frac{\pi}{6}} [\sin 4x + \sin 2x] dx$

$= -\frac{1}{2} \left[\frac{1}{4} \cos 4x \right]_0^{\frac{\pi}{6}} + -\frac{1}{2} \left[\frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{6}}$

$= -\frac{1}{8} \cos 4x \Big|_0^{\frac{\pi}{6}} - \frac{1}{4} \cos 2x \Big|_0^{\frac{\pi}{6}}$

(11) (d) (ii) $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$ *ratio formula often messed up!*

$P(6p, 3p^2)$
 $B(0, -3p^2)$

$m:n = 1:2$

$A = \left(\frac{1 \times 0 + 2 \times 6p}{3}, \frac{-3p^2 + 6p^2}{3} \right)$
 $A(4p, p^2)$ (1)

So

$x = 4p \Rightarrow p = \frac{x}{4}$
 $y = p^2$

Many students found 'A' but could not find locus equation.

So $y = \frac{x^2}{16}$

$\Rightarrow x^2 = 16y$ is the locus. (1)

(e) (i) $T = H + Be^{-kt} \Rightarrow Be^{-kt} = T - H$
 $\frac{dT}{dt} = 0 - Bke^{-kt}$
 $= -k(T - H)$ (1)

(ii) $H = 180$

When $t = 0, T = 10$

$10 = 180 + Be^{-kt}$

$10 - 180 = B$

$B = -170$

So $T = 180 - 170e^{-kt}$

data $t = 30$
 $T = 50$

*Well answered.
 Using the method shown.
 Some boys wanted to
 integrate the $\frac{dT}{dt}$ and
 got into a lot of
 bother.*

$$\text{So } 50 = 180 - 170e^{-30k}$$

$$-130 = -170e^{-30k}$$

$$\left(\frac{13}{17}\right) = e^{-30k}$$

$$\ln\left(\frac{13}{17}\right) = \ln e^{-30k}$$

$$= -30k$$

$$k = \frac{\ln\left(\frac{13}{17}\right)}{-30} = 0.008942$$

Lucky k either as exact or approx will give the correct exact or approx. Final answer.

$$-0.008942t$$

$$\text{So } T = 180 - 170e^{-0.008942t}$$

②

$$\text{(iii) } 150 = 180 - 170e^{-0.008942t}$$

$$-30 = -170e^{-0.008942t}$$

$$\left(\frac{3}{17}\right) = e^{-0.008942t}$$

$$\ln\left(\frac{3}{17}\right) = \ln e^{-0.008942t}$$

$$= -0.008942t$$

$$t = \frac{\ln\left(\frac{3}{17}\right)}{-0.008942} \doteq 194 \text{ mins}$$

3 hrs 14 mins

Well answered.

②

$$\begin{aligned}
& \left(-\frac{1}{8} \cos \frac{4\pi}{6} + \frac{1}{8} \cos 0 \right) - \left(\frac{1}{4} \cos \frac{2\pi}{6} - \frac{1}{4} \cos 0 \right) \\
&= -\frac{1}{8} \cos \frac{2\pi}{3} + \frac{1}{8} - \left(\frac{1}{4} \cos \frac{\pi}{3} - \frac{1}{4} \right) \\
&= -\frac{1}{8} \cos 120^\circ + \frac{1}{8} - \frac{1}{4} \cos 60^\circ + \frac{1}{4} \\
&= -\frac{1}{8} \times -\frac{1}{2} + \frac{1}{8} - \frac{1}{4} \cdot \frac{1}{2} + \frac{1}{4} \\
&= \frac{1}{16} + \frac{1}{8} - \frac{1}{8} + \frac{1}{4} = \frac{1}{16} + \frac{1}{4} = \frac{5}{16} //
\end{aligned}$$

Many students did not use hint from (c)(i) or used it badly. The $\frac{1}{2}$ out the front was there in most answers. Those that made the correction that (i) can help in (ii), most students worked out the answer. (0.3125)

(2)

(d) $P(6p, 3p^2)$

$x^2 = 12y$

$y - 3p^2 = px - 6p^2$
 $y = px - 3p^2 \quad (2)$

(i) $x^2 = 12y$

$y = \frac{x^2}{12}$

$y' = \frac{2x}{12} = \frac{x}{6}$

at $x = 6p$ $m = \frac{6p}{6} = p$

$(y - 3p^2) = p(x - 6p)$

well answered.

(ii) Cuts y axis at B.

$x = 0 \quad y = -3p^2$

$B(0, -3p^2) \quad (1)$

$P(6p, 3p^2)$

easy to find

12. (a) (i) $\cos x - \sqrt{3} \sin x = 1 \quad 0 \leq x \leq 2\pi$

$$\cos x - \sqrt{3} \sin x = R \cos(x + \alpha)$$

$$\cos x - \sqrt{3} \sin x = R \cos x \cos \alpha - R \sin x \sin \alpha$$

$$\Rightarrow R = \sqrt{1+3} = 2$$

$$\text{Then } 2 \cos \alpha = 1 \quad \text{and } 2 \sin \alpha = \sqrt{3}$$

$$\cos \alpha = \frac{1}{2}$$

$$\sin \alpha = \frac{\sqrt{3}}{2}$$

$$\therefore \alpha = \frac{\pi}{3}$$

$$\text{Then } 2 \cos\left(x + \frac{\pi}{3}\right) = 1$$

$$\cos\left(x + \frac{\pi}{3}\right) = \frac{1}{2}$$

$$x + \frac{\pi}{3} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \dots$$

$$x = 0, \frac{4\pi}{3}, 2\pi, \dots$$

(ii) General Soln: $x = 2n\pi$ or $2n\pi \pm \frac{4\pi}{3}$

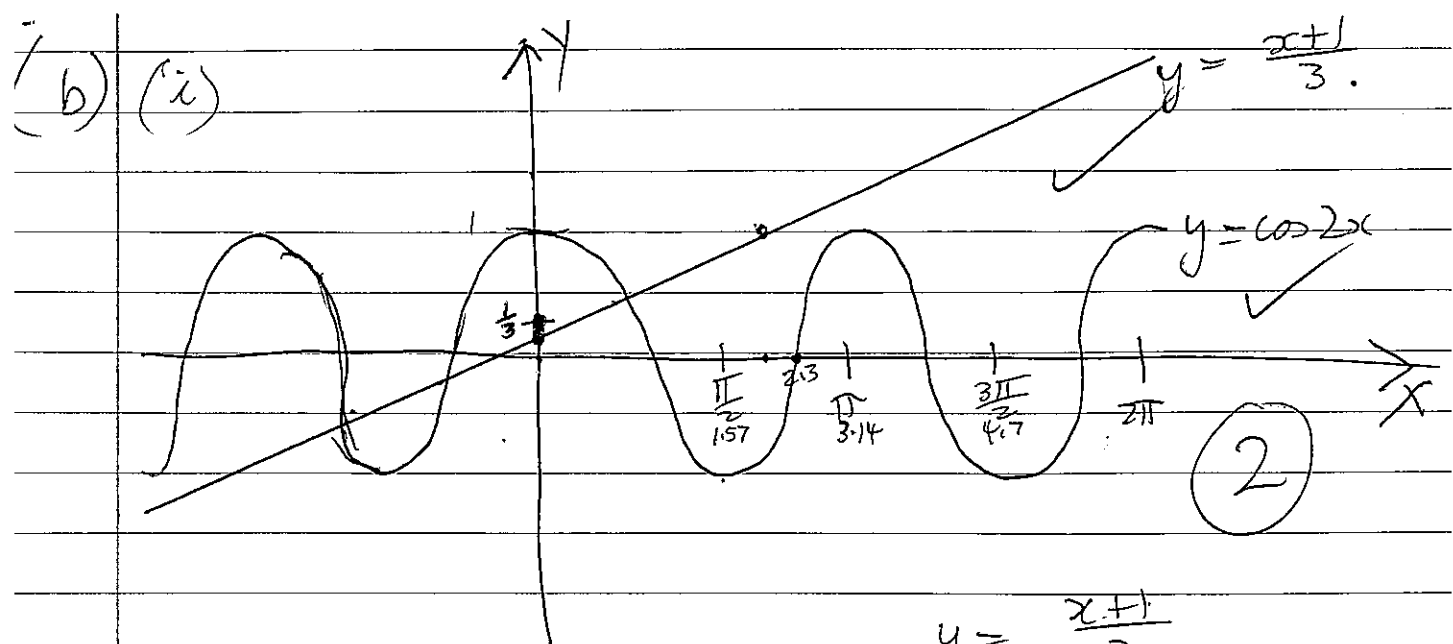
$$\text{or } x = -\frac{\pi}{3} + 2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$$

$$\text{or } x = -\frac{2\pi}{3} + 2k\pi \text{ or } 2k\pi, k \in \mathbb{Z}$$

(i) Most students realised they needed the auxiliary angle method. Common errors included:

- evaluating tan incorrectly and having $1/\sqrt{3}$
- not finding all the solutions in the given domain.

(ii) Some had the incorrect formula.



(ii) 3 solutions ✓ (1)

x	0	2
y	$\frac{1}{3}$	1

(iii) $f(x) = 3\cos 2x - x - 1$

Let $f(0.5) = 1.6209 - 0.5 - 1$
 $f(0.5) = 0.1209069$

Also $f'(x) = -6\sin 2x - 1$ ✓
 $f'(x) = -6.0488$

Then $x_{n+1} = x_1 - \frac{f(x_1)}{f'(x_1)}$
 $= 0.5 - \frac{0.1209069}{-6.0488259}$ ✓ (3)

$= 0.5199884907$
 $x_{n+1} = 0.520$ to 3dp ✓

An inaccurate graph resulted in the wrong number of solutions.
 Using Newton's Method done well on the whole. Some students did not use the given starting value and so were incorrect.

$$(c) \int_0^{\pi/4} \sin^2 2x \, dx$$

$$\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$$

$$\sin^2 2x = \frac{1}{2}(1 - \cos 4x)$$

$$= \frac{1}{2} \int_0^{\pi/4} (1 - \cos 4x) \, dx$$

$$= \frac{1}{2} \left[x - \frac{\sin 4x}{4} \right]_0^{\pi/4}$$

$$= \frac{1}{2} \left[\left(\frac{\pi}{4} - 0 \right) - \left(0 - 0 \right) \right]$$

$$= \frac{\pi}{8}$$

The most common mistake was not using the double angle formula correctly.

(3)

$$(d) \quad x = \frac{-5}{(x+2)^3}$$

When $t=0$, $x = x_{cm}$.

$t=0$, $v=2$, $x=0$.

$$\frac{d(\frac{1}{2}v^2)}{dx} = \frac{-5}{(x+2)^3}$$

$$\frac{1}{2}v^2 = -5 \int \frac{1}{(x+2)^3} \, dx$$

$$v^2 = \frac{-10(x+2)^{-2}}{-2} + C$$

$$v^2 = \frac{5}{(x+2)^2} + C$$

When $x=0$, $v=2 \Rightarrow 4 = \frac{5}{4} + C$
 $C = 2\frac{3}{4} = \frac{11}{4}$

(d) (cont)

$$v^2 = \frac{5}{(x+2)^2} + \frac{11}{4}$$

$v > 0$ since v can never be 0

Also when $t=0$, $v=2$ and

$\frac{d^2x}{dt^2} < 0 \Rightarrow$ decreasing speed.

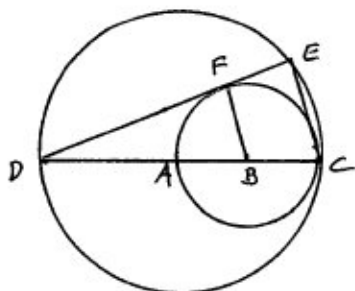
$$\therefore v = \sqrt{\frac{5}{(x+2)^2} + \frac{11}{4}}$$

The most common error was to differentiate the given function in terms of t .

Half a mark was deducted if no statement about sign of v was included.

✓ (3)

(a)



$FB \perp DF$ (radius $\perp$ tangent at point of contact)

$$DB^2 = FB^2 + DF^2 \text{ (Pythagoras' Theorem)}$$

$$\therefore 9^2 = 3^2 + DF^2$$

$$\therefore DF^2 = 72$$

$$\therefore DF = 6\sqrt{2}$$

$EC \perp DE$ (DC is a diameter
 $\therefore \angle DEC = 90^\circ$, angle in a semicircle)

$\therefore \triangle DBF \parallel \triangle DCE$ (equiangular)

as $\angle D$ is common

$$\angle DFB = \angle DEC = 90^\circ \text{ (as indicated above)}$$

$$\therefore \frac{DE}{DF} = \frac{DC}{CB} \text{ (corresponding sides in similar triangles)}$$

$$\therefore \frac{DE}{6\sqrt{2}} = \frac{12}{9}$$

$$\therefore DE = \frac{12 \times 6\sqrt{2}}{9}$$

$$= 8\sqrt{2} \quad (4)$$

There seemed to be a reluctance to give reasons for geometrical conclusions

$$(b) (i) K! + K \times K!$$

$$= K! (1 + K)$$

$$= (K+1)! \quad (1)$$

Done well although some stopped at $K!(1+K)$

0	0.5	1	Mean
6	18	138	0.91

$$(ii) S(n) \equiv 1 \times 1! + 2 \times 2! + \dots + n \times n! = (n+1)! - 1$$

Show $S(1)$ is true

$$\text{i.e. } 1 \times 1! = 2! - 1$$

$$\text{LHS} = 1$$

$$\text{RHS} = 2 - 1$$

$$= 1$$

$\therefore S(1)$ is true

Assume $S(k)$ is true

$$\text{i.e. } 1 \times 1! + 2 \times 2! + \dots + k \times k! = (k+1)! - 1$$

Show $S(k+1)$ is true

$$\text{i.e. } 1 \times 1! + 2 \times 2! + \dots + k \times k! + (k+1) \times (k+1)! = (k+2)! - 1$$

$$\text{LHS} = (k+1)! - 1 + (k+1) \times (k+1)!$$

$$= (k+1)! (1 + k+1) - 1$$

$$= (k+1)! (k+2) - 1$$

$$= (k+2)! - 1$$

$$= \text{RHS}$$

$\therefore$ If $S(k)$ is true, $S(k+1)$ is true

$S(1)$ is true and, if $S(k)$ is true,

$S(k+1)$ is true

$\therefore$ By the process of Mathematical Induction, $S(n)$ is true for all integral $n \geq 1$. (3)

0	0.5	1	1.5	2	2.5	3	3.5	4	Mean
7	6	11	14	25	8	20	22	49	2.70

Most students demonstrated an understanding of the process of Mathematical Induction. However, many statements were sloppy. For example, "Assume $n=k$ " rather than "Assume the statement is true if $n=k$ " or, having defined the statement as $S(n)$ as above, "Assume that $S(k)$ is true". Many concluding statements were also sloppy.

0	0.5	1	1.5	2	2.5	3	Mean
2	8	2	3	0	4	143	2.77

(c) (i) If $x = 3 + 3 \sin \theta$,
 $dx = 3 \cos \theta$
 and $\sin \theta = \frac{x-3}{3}$.

$$\begin{aligned} & \int \sqrt{x(6-x)} \, dx \\ &= \int \sqrt{(3+3\sin\theta)(6-3-3\sin\theta)} \cdot 3\cos\theta \, d\theta \\ &= \int 3\sqrt{(1-\sin^2\theta)} \cdot 3\cos\theta \, d\theta \\ &= 9 \int \cos^2\theta \, d\theta \end{aligned}$$

NOTE: ORIGINAL INTEGRAL IS POSITIVE

$$= 9 \int \frac{\cos 2\theta + 1}{2} \, d\theta$$

$$\begin{aligned} &= \frac{9}{2} \left[\frac{\sin 2\theta}{2} + \theta \right] + C \\ &= \frac{9}{2} \left[\sin\theta \cos\theta + \theta \right] + C \\ &= \frac{9}{2} \left[\frac{x-3}{3} \sqrt{1-\frac{(x-3)^2}{9}} + \sin^{-1} \frac{x-3}{3} \right] + C \\ &= \frac{9}{2} \left[\frac{x-3}{3} \sqrt{\frac{6x-x^2}{9}} + \sin^{-1} \frac{x-3}{3} \right] + C \\ &= \frac{1}{2} \left[(x-3)\sqrt{x(6-x)} + 9 \sin^{-1} \frac{x-3}{3} \right] + C \end{aligned}$$

(4)

Many students found this integration challenging. Some left the integral at the form $\frac{9}{2} [\sin\theta \cos\theta + \theta] + C$, or equivalent rather than returning to an expression in terms of x .

0	0.5	1	1.5	2	2.5	3	3.5	4	Mean
3	5	11	4	20	15	25	24	55	2.93

(ii) $V = \pi \int_0^6 \left(\sqrt[4]{x(6-x)} \right)^2 \, dx$

$$\begin{aligned} &= \pi \int_0^6 \sqrt{x(6-x)} \, dx \\ &= \pi \times \frac{1}{2} \left[(x-3)\sqrt{x(6-x)} + 9 \sin^{-1} \left(\frac{x-3}{3} \right) \right]_0^6 \\ &= \frac{\pi}{2} \left\{ \left[9 \sin^{-1} 1 \right] - \left[9 \sin^{-1} (-1) \right] \right\} \\ &= \frac{\pi}{2} \left\{ 9 \cdot \frac{\pi}{2} - 9 \left(-\frac{\pi}{2} \right) \right\} \\ &= \frac{9\pi^2}{2} \end{aligned}$$

(3)

Most who progressed through (c) (i) found the appropriate volume.

0	0.5	1	1.5	2	2.5	3	Mean
15	10	21	4	24	23	65	2.05

14(a)

$$\left| \frac{dV}{dt} = \frac{2\pi}{5} \right| \text{ (given)}$$

$$S = \pi r^2$$

$$V = \frac{1}{3} \pi r^2 h.$$

$$\left| \frac{dS}{dr} = 2\pi r \right|$$

$$d \frac{h}{10} = \frac{r}{8} \text{ (SIMILARITY)}$$

$$\therefore V = \frac{5}{12} \pi r^3$$

$$\left| \frac{dV}{dr} = \frac{5}{4} \pi r^2 \right|$$

$$\text{Now } \frac{dS}{dt} = \frac{dS}{dr} \times \frac{dr}{dt} \times \frac{dV}{dt}.$$

$$= 2\pi r \times \frac{4}{5\pi r^2} \times \frac{2\pi}{5}$$

$$= \frac{16\pi}{25r}$$

$$\text{When } h=4$$

$$r = \frac{16}{5}$$

$$= \frac{16}{25} \times \pi \times \frac{5}{16}$$

$$= \frac{\pi}{5} \text{ cm/min.}$$

COMMENT not particularly well done.

many students treated h as a constant in the differentiation

of $\frac{dV}{dr}$.

(b) (i) let $y = 0$.

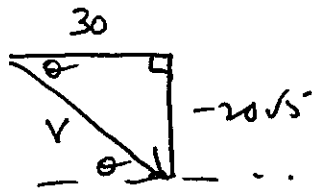
$$100 - 5t^2 = 0$$

$$5t^2 = 100$$

$$t^2 = 20$$

$$t = 2\sqrt{5} \text{ secs.}$$

$$(ii) \quad \dot{x} = 30 \quad \dot{y} = -10t \\ = -20\sqrt{5}$$



$$V^2 = 900 + 2000 \\ = \sqrt{2900}$$

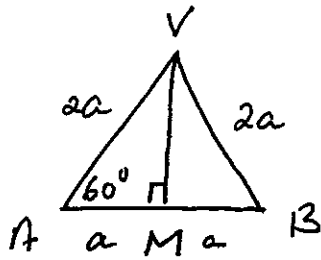
$$V = 10\sqrt{29} \text{ m/s.}$$

$$\theta = \tan^{-1} \frac{2\sqrt{5}}{3}$$

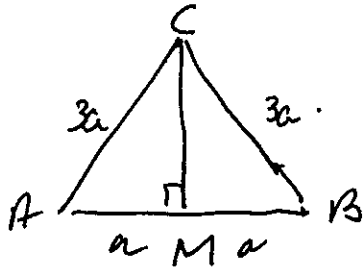
COMMENT Common error was

to let $y = -100$. Generally will do.

(C).



$$\begin{aligned} VM &= 2a \sin 60^\circ \\ &= 2a \frac{\sqrt{3}}{2} \\ \underline{VM} &= \underline{a\sqrt{3}}. \end{aligned}$$



$$\begin{aligned} CM^2 &= (3a)^2 - a^2 \\ &= 8a^2 \\ \therefore \underline{CM} &= \underline{\sqrt{8a^2}}. \end{aligned}$$

$$\begin{aligned} \cos \theta &= \frac{CM^2 + VM^2 - VC^2}{2 \times a\sqrt{3} \times 2\sqrt{2}a} \\ &= \frac{8a^2 + 3a^2 - 5a^2}{4\sqrt{6}a^2} \\ &= \frac{6a^2}{4\sqrt{6}a^2} \end{aligned}$$

$$\therefore \boxed{\cos \theta = \frac{\sqrt{6}}{4}}.$$

COMMENT: Quite well done.

$$\begin{aligned} (11) \quad VO &= VM \sin \theta. \\ &= a\sqrt{3} \times \frac{\sqrt{10}}{4} \\ &= \underline{\underline{a\sqrt{30}}}. \end{aligned}$$

$$\begin{aligned} \sin^2 \theta &= 1 - \frac{6}{16} \\ &= \frac{10}{16} \\ \therefore \sin \theta &= \frac{\sqrt{10}}{4} \end{aligned}$$

COMMENT:

many students unable to find $\sin \theta$.

$$\begin{aligned}
 \text{(III)} \quad \cos \angle B &= \frac{(2a)^2 + (3a)^2 - (\sqrt{5}a)^2}{2 \times 2a \times 3a} \\
 &= \frac{4a^2 + 9a^2 - 5a^2}{12a^2} \\
 &= \frac{2}{3}
 \end{aligned}$$

$$\begin{aligned}
 \therefore VP^2 &= VB^2 + (ra)^2 - 2 \times 2a \times ra \times \frac{2}{3} \\
 &= 4a^2 + r^2 a^2 - \frac{8ra^2}{3} \\
 &= \frac{a^2}{3} [12 + 3r^2 - 8r]
 \end{aligned}$$

COMMENT Very few students were able to obtain this answer.

The common error was to assume

$\triangle COP \parallel \triangle CMB$. Hence finding an expression for OP then using Pythagoras to obtain VP^2 . (this was not given marks)

$$\text{(IV)} \quad VP = a \sqrt{\frac{12 + 3r^2 - 8r}{3}}$$

$$\begin{aligned}
 \therefore \sin \phi &= \frac{\frac{a\sqrt{30}}{4}}{a \sqrt{\frac{12 + 3r^2 - 8r}{3}}} \\
 &= \frac{\sqrt{90}}{4 \sqrt{12 + 3r^2 - 8r}} \\
 &= \frac{\sqrt{45}}{\sqrt{8(12 + 3r^2 - 8r)}}
 \end{aligned}$$

COMMENT-

most obtained a mark, attending for previous error in VO.

(V). Best done by recognising

that $\sin \phi$ is maximised
by $3r^2 - 8r + 12$ being a minimum
this occurs when $6r - 8 = 0$
 $r = 4/3$.

$$\begin{aligned} \text{ii. } \sin \phi &= \sqrt{\frac{45}{8(3r^2 - 8r + 12)}} \quad \text{where } r = \frac{4}{3} \\ &= \sqrt{\frac{45 \times 9}{8 \times 60}} \\ &= \frac{3\sqrt{6}}{8} \end{aligned}$$

$$\therefore \phi = \sin^{-1} \frac{3\sqrt{6}}{8}$$

COMMENT:

many students were
able to obtain a mark or two.

many maximised $\sin \phi$ by
calculus. Few saw the easier
approach.